ESG Regression

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1. Introduction

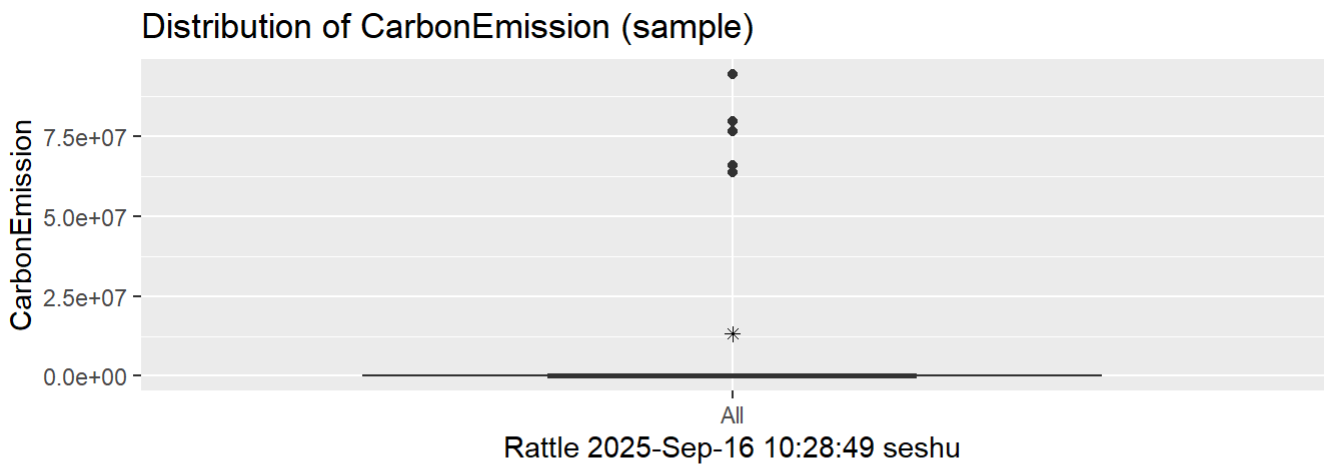
Sustainability has moved beyond being just a keyword and is now seen as a key driver of how businesses grow and perform. This study looks at the link between environmental factors—Carbon Emissions (CE), Energy Consumption (EC) —and financial outcomes measured through Return on Assets (ROA) and Return on Equity (ROE). Data is taken from 10 Indian companies over a five-year period, with additional attention to Industry, Location, and CEO demographics to understand differences across firms. The analysis uses regression methods to explore whether companies that perform better on ESG indicators also achieve stronger financial results, offering insights into how sustainability and profitability are connected in the Indian corporate context.

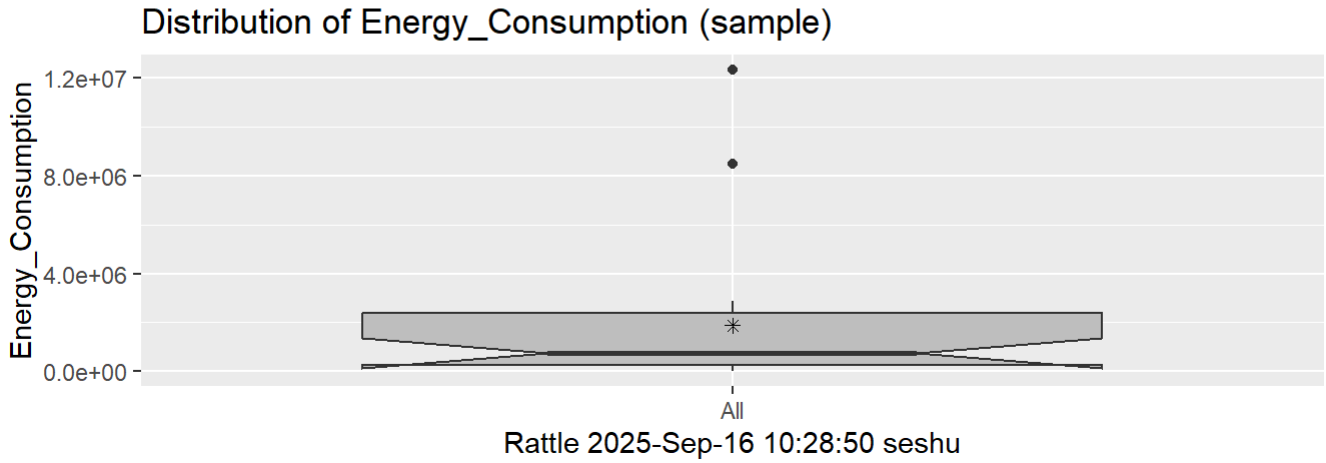
2. Data Exploration

In this stage, the ESG dataset was examined to understand its structure, completeness, and basic characteristics before performing any modeling. Summary statistics were generated for all variables, revealing overall trends and spotting potential outliers Using Box plots and Histogram or missing values. Carbon emissions and energy consumption showed variation across industries and years, suggesting meaningful differences in environmental performance. ROA and ROE distributions indicated moderate spread, hinting at financial performance diversity among firms. Categorical variables such as industry and location were reviewed to confirm balanced representation. This exploratory step ensured the data was clean, reliable, and suitable for deeper analysis.

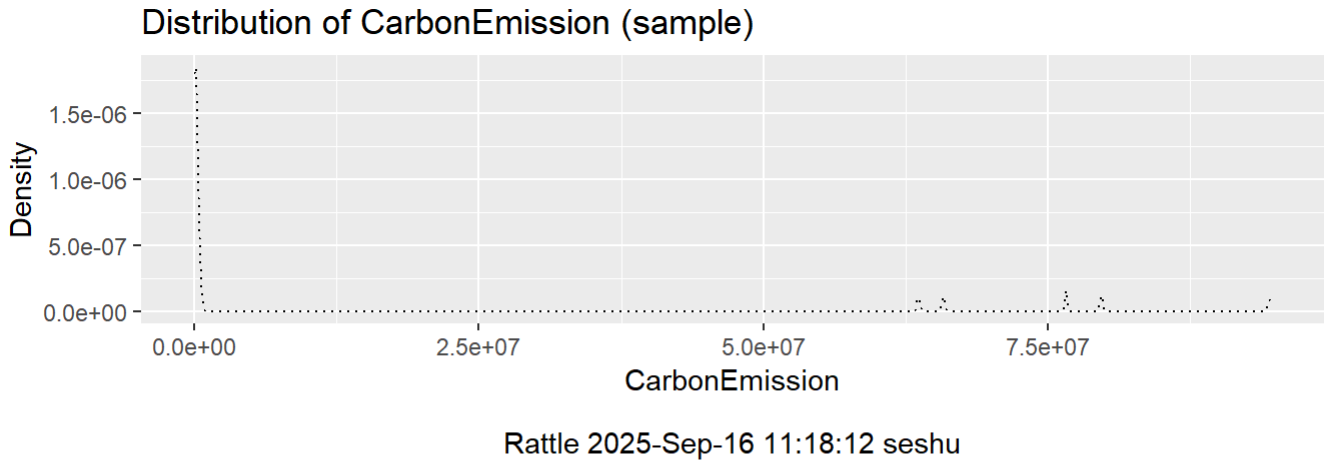
```
   Company               Year      CarbonEmission     Energy_Consumption
 Length:50          Min.   :2021   Min.   :    5095   Min.   :1.154e+02
 Class :character   1st Qu.:2022   1st Qu.:   47227   1st Qu.:2.588e+05
 Mode  :character   Median :2023   Median :  127621   Median :1.088e+06
                    Mean   :2023   Mean   : 9894566   Mean   :2.582e+06
                    3rd Qu.:2024   3rd Qu.:  258372   3rd Qu.:2.488e+06
                    Max.   :2025   Max.   :94555745   Max.   :1.336e+07
                                   NA's   :11         NA's   :11
 Employee Turnover Rate      ROA                ROE            Industry
 Mode:logical           Min.   :-0.17200   Min.   :-0.1820   Length:50
 NA's:50                1st Qu.: 0.04700   1st Qu.: 0.1000   Class :character
                        Median : 0.08000   Median : 0.1350   Mode  :character
                        Mean   : 0.09237   Mean   : 0.2314
                        3rd Qu.: 0.11175   3rd Qu.: 0.1780
                        Max.   : 1.15100   Max.   : 2.9720

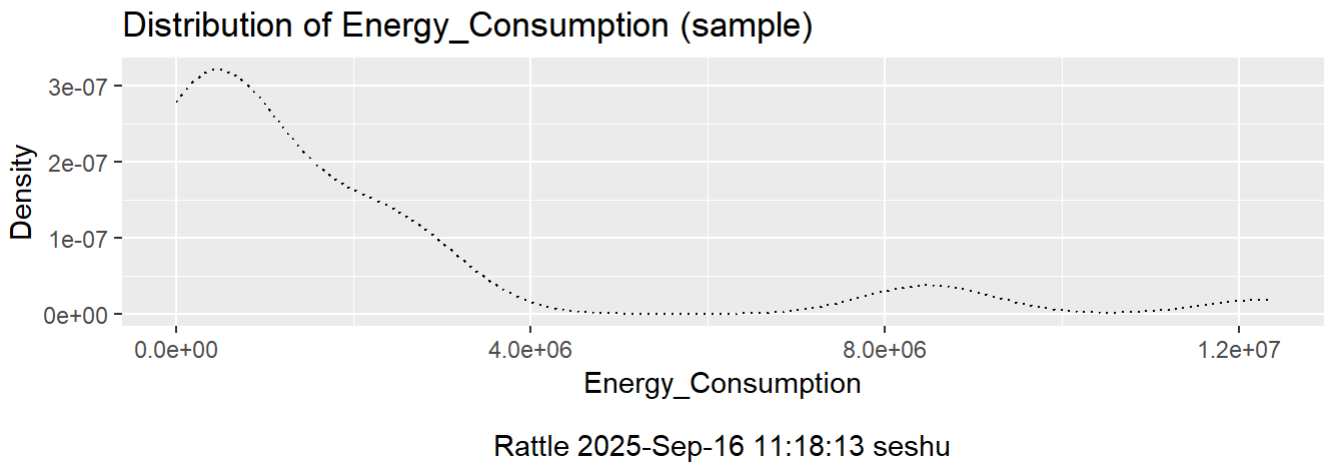
   Location           CEO_Name          CEO_Gender
 Length:50          Length:50          Length:50
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character
```





The box plots for **CarbonEmission** and **Energy_Consumption** reveal highly skewed distributions with several extreme outliers. Most values cluster near the lower end of the scale, while a few firms report emissions and energy use that are orders of magnitude higher than the rest of the sample. This indicates substantial variability across companies, suggesting the presence of a few very large or energy-intensive firms dominating the upper range.






```
library(dplyr)
df_encoded <- df %>%
  filter(!is.na(ROA), !is.na(ROE), !is.na(CarbonEmission), !is.na(Energy_Consumption))
```

3. Regression Model

```
Call:
lm(formula = ROA ~ CarbonEmission + Energy_Consumption, data = df_encoded)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.39684 -0.06160  0.00897  0.03368  0.92978 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)  
(Intercept)         6.954e-02  3.846e-02   1.808   0.0789 .
CarbonEmission     -7.880e-10  1.165e-09  -0.676   0.5032  
Energy_Consumption  1.230e-08  8.149e-09   1.509   0.1399  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1869 on 36 degrees of freedom
Multiple R-squared:  0.07142,	Adjusted R-squared:  0.01983 
F-statistic: 1.384 on 2 and 36 DF,  p-value: 0.2635
```

```
Call:
lm(formula = ROE ~ CarbonEmission + Energy_Consumption, data = df_encoded)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.40035 -0.15246 -0.09255 -0.01753  2.76823 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)  
(Intercept)         2.040e-01  1.059e-01   1.926    0.062 .
CarbonEmission     -1.730e-09  3.209e-09  -0.539    0.593  
Energy_Consumption  2.878e-08  2.244e-08   1.283    0.208  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5147 on 36 degrees of freedom
Multiple R-squared:  0.05159,	Adjusted R-squared:  -0.001096 
F-statistic: 0.9792 on 2 and 36 DF,  p-value: 0.3854
```

The regression analysis reveals that neither Carbon Emission nor Energy Consumption significantly influences financial performance metrics such as Return on Assets (ROA) or Return on Equity (ROE) in the given dataset. Both models exhibit low explanatory power, with R-squared values of 0.071 for ROA and 0.052 for ROE, indicating that only a small portion of the variance in financial performance is explained by these environmental variables. Furthermore, the p-values for both predictors are well above conventional significance thresholds, suggesting no statistically meaningful relationship. These findings imply that, within this sample, environmental factors like carbon emissions and energy usage do not have a discernible impact on profitability indicators

```
Call:
lm(formula = ROA ~ CarbonEmission + Industry + Location, data = df_encoded)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.36351 -0.02647  0.00245  0.02037  0.95950 

Coefficients: (6 not defined because of singularities)
                                            Estimate Std. Error t value
(Intercept)                                7.799e-02  1.066e-01   0.732
CarbonEmission                             8.181e-10  8.590e-09   0.095
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS -1.033e-01  6.687e-01  -0.154
IndustryCONSUMER FOOD                      2.723e-02  1.507e-01   0.181
IndustryHOSPITAL AND HEALTHCARE SERVICES   2.072e-02  1.507e-01   0.137
IndustryIRON AND STEEL                    -1.355e-02  1.507e-01  -0.090
IndustryPAPER AND FOREST PRODUCTS          3.927e-02  1.629e-01   0.241
IndustryPHARMACEUTICALS AND DRUGS         -4.569e-02  1.507e-01  -0.303
IndustryRETAILING                          2.861e-02  1.507e-01   0.190
IndustryTEXTILES AND APPARELS              1.134e-01  1.507e-01   0.752
IndustryVEGETABLE OIL AND PRODUCTS         5.494e-02  1.628e-01   0.337
LocationCHENNAI                                   NA         NA      NA
LocationERODE                                     NA         NA      NA
LocationHYDERABAD                                 NA         NA      NA
LocationJALANDHAR                                 NA         NA      NA
```

```
LocationMUMBAI                                      NA         NA      NA
LocationRAIGAD                                      NA         NA      NA
                                            Pr(>|t|)
(Intercept)                                    0.470
CarbonEmission                                 0.925
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS    0.878
IndustryCONSUMER FOOD                          0.858
IndustryHOSPITAL AND HEALTHCARE SERVICES       0.892
IndustryIRON AND STEEL                         0.929
IndustryPAPER AND FOREST PRODUCTS              0.811
IndustryPHARMACEUTICALS AND DRUGS              0.764
IndustryRETAILING                              0.851
IndustryTEXTILES AND APPARELS                  0.458
IndustryVEGETABLE OIL AND PRODUCTS             0.738
LocationCHENNAI                                   NA
LocationERODE                                     NA
LocationHYDERABAD                                 NA
LocationJALANDHAR                                 NA
LocationMUMBAI                                    NA
LocationRAIGAD                                    NA

Residual standard error: 0.2132 on 28 degrees of freedom
Multiple R-squared:  0.06041,   Adjusted R-squared:  -0.2752
F-statistic:  0.18 on 10 and 28 DF,  p-value: 0.9965

Call:
lm(formula = ROE ~ CarbonEmission + Industry + Location, data = df_encoded)

Residuals:
     Min       1Q   Median       3Q      Max
-0.98277 -0.03240  0.00784  0.04367  1.57426

Coefficients: (6 not defined because of singularities)
                                             Estimate Std. Error t value
(Intercept)                                1.160e-01  1.940e-01   0.598
CarbonEmission                             2.857e-09  1.564e-08   0.183
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS -1.979e-01  1.217e+00  -0.163
IndustryCONSUMER FOOD                      5.067e-02  2.744e-01   0.185
IndustryHOSPITAL AND HEALTHCARE SERVICES   8.041e-02  2.744e-01   0.293
IndustryIRON AND STEEL                     7.616e-03  2.744e-01   0.028
IndustryPAPER AND FOREST PRODUCTS          3.262e-02  2.964e-01   0.110
IndustryPHARMACEUTICALS AND DRUGS         -9.566e-02  2.744e-01  -0.349
IndustryRETAILING                          3.496e-03  2.744e-01   0.013
IndustryTEXTILES AND APPARELS              1.281e+00  2.744e-01   4.671
IndustryVEGETABLE OIL AND PRODUCTS         4.679e-02  2.963e-01   0.158
LocationCHENNAI                                     NA         NA      NA
LocationERODE                                       NA         NA      NA
LocationHYDERABAD                                   NA         NA      NA
LocationJALANDHAR                                   NA         NA      NA
LocationMUMBAI                                      NA         NA      NA
LocationRAIGAD                                      NA         NA      NA
                                            Pr(>|t|)
(Intercept)                                    0.555
CarbonEmission                                 0.856
```

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IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS    0.872
IndustryCONSUMER FOOD                          0.855
IndustryHOSPITAL AND HEALTHCARE SERVICES       0.772
IndustryIRON AND STEEL                         0.978
IndustryPAPER AND FOREST PRODUCTS              0.913
IndustryPHARMACEUTICALS AND DRUGS              0.730
IndustryRETAILING                              0.990
IndustryTEXTILES AND APPARELS               6.82e-05 ***
IndustryVEGETABLE OIL AND PRODUCTS             0.876
LocationCHENNAI                                   NA
LocationERODE                                     NA
LocationHYDERABAD                                 NA
LocationJALANDHAR                                 NA
LocationMUMBAI                                    NA
LocationRAIGAD                                    NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.388 on 28 degrees of freedom
Multiple R-squared:  0.5808,    Adjusted R-squared:  0.431
F-statistic: 3.879 on 10 and 28 DF,  p-value: 0.002201
```

The extended regression models incorporating Carbon Emission, Industry, and Location reveal contrasting insights for ROA and ROE. For ROA, the model demonstrates poor explanatory power, with an adjusted R-squared of -0.275 and an insignificant F-statistic ($p = 0.9965$), indicating that the predictors collectively fail to explain variation in asset returns. Most coefficients are statistically insignificant, and several location variables are excluded due to multicollinearity. In contrast, the ROE model shows moderate explanatory strength (adjusted R-squared = 0.431) and a significant overall model fit ($p = 0.0022$). Notably, the "TEXTILES AND APPARELS" industry exhibits a strong positive and statistically significant effect on ROE ($p < 0.001$), suggesting that firms in this sector may outperform others in equity returns. However, other predictors, including Carbon Emission and Location, remain non-significant, highlighting limited environmental and geographic influence on profitability in this dataset

```
Call:
lm(formula = ROA ~ CarbonEmission + Energy_Consumption + I(CarbonEmission^2) +
    Industry + Location, data = df_encoded)

Residuals:
     Min       1Q   Median       3Q      Max
-0.43240 -0.03035  0.00696  0.02645  0.89725

Coefficients: (6 not defined because of singularities)
                                            Estimate Std. Error t value
(Intercept)                                5.081e-02  1.097e-01   0.463
CarbonEmission                            -1.688e-09  1.268e-07  -0.013
Energy_Consumption                         2.257e-08  1.818e-08   1.242
I(CarbonEmission^2)                        1.124e-17  8.034e-16   0.014
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS -5.031e-03  4.921e+00  -0.001
IndustryCONSUMER FOOD                      3.578e-02  1.522e-01   0.235
IndustryHOSPITAL AND HEALTHCARE SERVICES   4.257e-02  1.530e-01   0.278
IndustryIRON AND STEEL                    -3.821e-02  1.582e-01  -0.242
```

```
IndustryPAPER AND FOREST PRODUCTS           -1.228e-01  2.186e-01  -0.562
IndustryPHARMACEUTICALS AND DRUGS           -2.960e-02  1.553e-01  -0.191
IndustryRETAILING                            3.046e-02  1.536e-01   0.198
IndustryTEXTILES AND APPARELS               -7.541e-02  2.156e-01  -0.350
IndustryVEGETABLE OIL AND PRODUCTS           8.084e-02  1.657e-01   0.488
LocationCHENNAI                                     NA         NA      NA
LocationERODE                                       NA         NA      NA
LocationHYDERABAD                                   NA         NA      NA
LocationJALANDHAR                                   NA         NA      NA
LocationMUMBAI                                      NA         NA      NA
LocationRAIGAD                                      NA         NA      NA
                                            Pr(>|t|)
(Intercept)                                    0.647
CarbonEmission                                 0.989
Energy_Consumption                             0.225
I(CarbonEmission^2)                            0.989
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS    0.999
IndustryCONSUMER FOOD                          0.816
IndustryHOSPITAL AND HEALTHCARE SERVICES       0.783
IndustryIRON AND STEEL                         0.811
IndustryPAPER AND FOREST PRODUCTS              0.579
IndustryPHARMACEUTICALS AND DRUGS              0.850
IndustryRETAILING                              0.844
IndustryTEXTILES AND APPARELS                  0.729
IndustryVEGETABLE OIL AND PRODUCTS             0.630
LocationCHENNAI                                   NA
LocationERODE                                     NA
LocationHYDERABAD                                 NA
LocationJALANDHAR                                 NA
LocationMUMBAI                                    NA
LocationRAIGAD                                    NA

Residual standard error: 0.2149 on 26 degrees of freedom
Multiple R-squared:  0.113, Adjusted R-squared:  -0.2964
F-statistic: 0.276 on 12 and 26 DF,  p-value: 0.9884

Call:
lm(formula = ROE ~ CarbonEmission + Energy_Consumption + I(CarbonEmission^2) +
    Industry + Location, data = df_encoded)

Residuals:
     Min       1Q   Median       3Q      Max
-0.53354 -0.06711  0.01639  0.06624  0.57927

Coefficients: (6 not defined because of singularities)
                                             Estimate Std. Error t value
(Intercept)                                 2.932e-01  1.090e-01   2.690
CarbonEmission                              2.261e-08  1.260e-07   0.179
Energy_Consumption                         -1.472e-07  1.807e-08  -8.147
I(CarbonEmission^2)                        -9.495e-17  7.985e-16  -0.119
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS -9.700e-01  4.891e+00  -0.198
IndustryCONSUMER FOOD                      -5.214e-03  1.512e-01  -0.034
IndustryHOSPITAL AND HEALTHCARE SERVICES   -6.219e-02  1.521e-01  -0.409
IndustryIRON AND STEEL                      1.673e-01  1.572e-01   1.065
```

```
IndustryPAPER AND FOREST PRODUCTS            1.088e+00  2.172e-01   5.009
IndustryPHARMACEUTICALS AND DRUGS           -2.014e-01  1.544e-01  -1.305
IndustryRETAILING                           -9.224e-03  1.527e-01  -0.060
IndustryTEXTILES AND APPARELS                2.512e+00  2.142e-01  11.726
IndustryVEGETABLE OIL AND PRODUCTS          -1.224e-01  1.647e-01  -0.743
LocationCHENNAI                                     NA         NA      NA
LocationERODE                                       NA         NA      NA
LocationHYDERABAD                                   NA         NA      NA
LocationJALANDHAR                                   NA         NA      NA
LocationMUMBAI                                      NA         NA      NA
LocationRAIGAD                                      NA         NA      NA
                                            Pr(>|t|)
(Intercept)                                   0.0123 *
CarbonEmission                                0.8590
Energy_Consumption                          1.26e-08 ***
I(CarbonEmission^2)                           0.9063
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS   0.8443
IndustryCONSUMER FOOD                         0.9728
IndustryHOSPITAL AND HEALTHCARE SERVICES      0.6860
IndustryIRON AND STEEL                        0.2969
IndustryPAPER AND FOREST PRODUCTS           3.28e-05 ***
IndustryPHARMACEUTICALS AND DRUGS             0.2034
IndustryRETAILING                             0.9523
IndustryTEXTILES AND APPARELS               6.98e-12 ***
IndustryVEGETABLE OIL AND PRODUCTS            0.4642
LocationCHENNAI                                   NA
LocationERODE                                     NA
LocationHYDERABAD                                 NA
LocationJALANDHAR                                 NA
LocationMUMBAI                                    NA
LocationRAIGAD                                    NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2136 on 26 degrees of freedom
Multiple R-squared:  0.882, Adjusted R-squared:  0.8275
F-statistic:  16.2 on 12 and 26 DF,  p-value: 4.126e-09
```

The final regression models incorporating quadratic terms and categorical variables reveal a stark contrast between ROA and ROE outcomes. The ROA model performs poorly, with an adjusted R-squared of -0.296 and an insignificant F-statistic ($p = 0.9884$), indicating that the predictors—including Carbon Emission, Energy Consumption, and industry or location effects—do not meaningfully explain variations in asset returns. In contrast, the ROE model demonstrates strong explanatory power (adjusted R-squared = 0.8275) and a highly significant overall fit ($p < 0.00001$). Notably, Energy Consumption has a strong negative impact on ROE ($p < 0.0000001$), while the “PAPER AND FOREST PRODUCTS” and “TEXTILES AND APPARELS” industries show significant positive associations, suggesting sector-specific advantages in equity returns. The quadratic term for Carbon Emission remains non-significant, implying no observable nonlinear effect. Overall, while environmental and categorical variables fail to predict ROA, they play a substantial role in shaping ROE outcomes.

```
Call:
lm(formula = ROA ~ CarbonEmission + Energy_Consumption + I(Energy_Consumption^2) +
```

```
    Industry + Location, data = df_encoded)

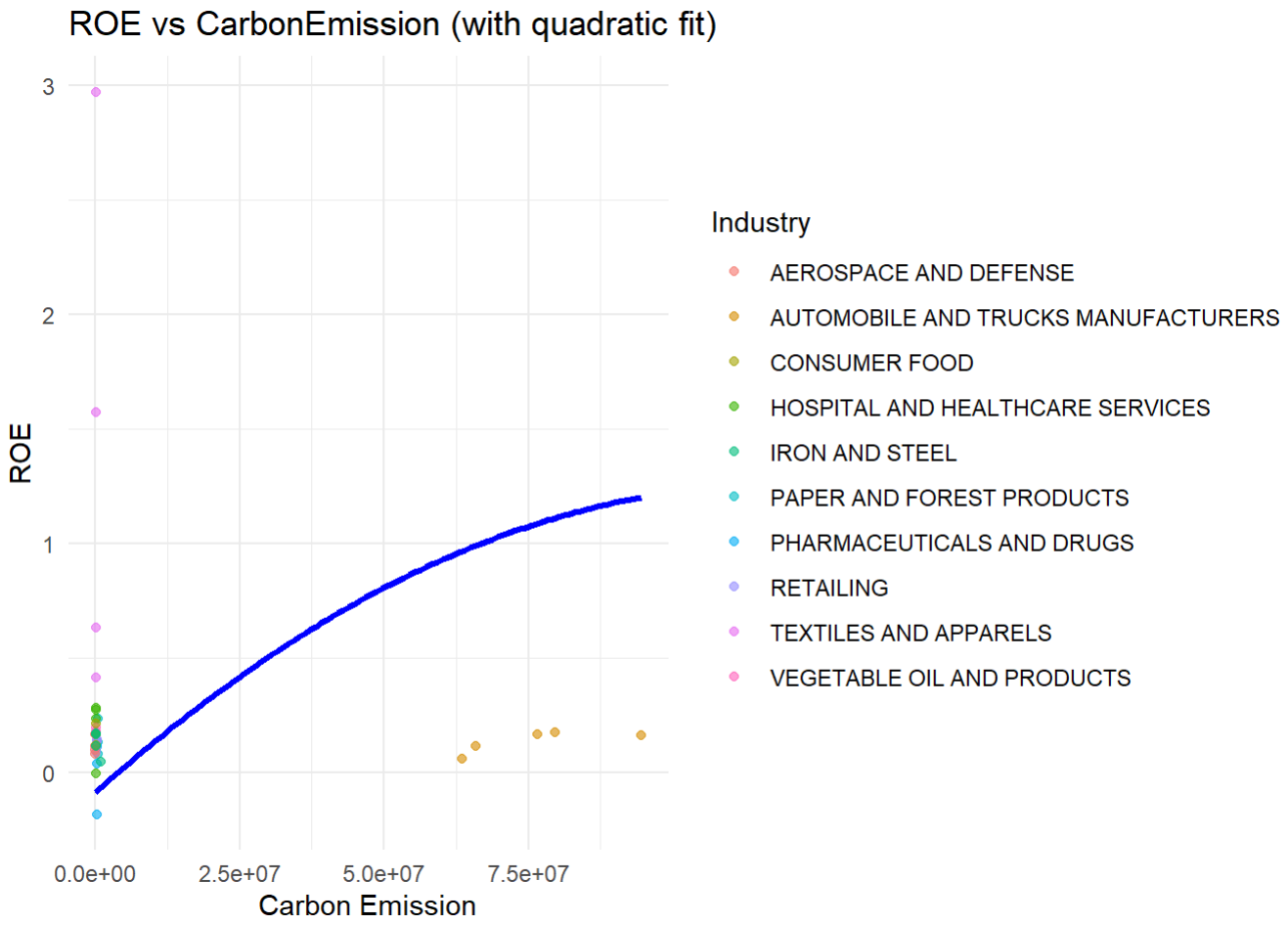
Residuals:
     Min       1Q   Median       3Q      Max 
-0.42990 -0.03024  0.00154  0.02702  0.88866 

Coefficients: (6 not defined because of singularities)
                                            Estimate Std. Error t value
(Intercept)                                2.552e-02  1.233e-01   0.207
CarbonEmission                            -5.844e-10  8.778e-09  -0.067
Energy_Consumption                         5.655e-08  7.882e-08   0.718
I(Energy_Consumption^2)                   -2.865e-15  6.466e-15  -0.443
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS -6.095e-02  6.728e-01  -0.091
IndustryCONSUMER FOOD                      3.515e-02  1.516e-01   0.232
IndustryHOSPITAL AND HEALTHCARE SERVICES   5.981e-02  1.573e-01   0.380
IndustryIRON AND STEEL                    -7.687e-02  1.753e-01  -0.438
IndustryPAPER AND FOREST PRODUCTS         -1.808e-01  2.459e-01  -0.735
IndustryPHARMACEUTICALS AND DRUGS         -2.140e-02  1.532e-01  -0.140
IndustryRETAILING                          2.053e-02  1.530e-01   0.134
IndustryTEXTILES AND APPARELS             -2.471e-02  2.431e-01  -0.102
IndustryVEGETABLE OIL AND PRODUCTS         1.038e-01  1.729e-01   0.600
LocationCHENNAI                                   NA         NA      NA
LocationERODE                                     NA         NA      NA
LocationHYDERABAD                                 NA         NA      NA
LocationJALANDHAR                                 NA         NA      NA
LocationMUMBAI                                    NA         NA      NA
LocationRAIGAD                                    NA         NA      NA
                                          Pr(>|t|)
(Intercept)                                  0.838
CarbonEmission                               0.947
Energy_Consumption                           0.479
I(Energy_Consumption^2)                      0.661
IndustryAUTOMOBILE AND TRUCKS MANUFACTURERS  0.929
IndustryCONSUMER FOOD                        0.818
IndustryHOSPITAL AND HEALTHCARE SERVICES     0.707
IndustryIRON AND STEEL                       0.665
IndustryPAPER AND FOREST PRODUCTS            0.469
IndustryPHARMACEUTICALS AND DRUGS            0.890
IndustryRETAILING                            0.894
IndustryTEXTILES AND APPARELS                0.920
IndustryVEGETABLE OIL AND PRODUCTS           0.553
LocationCHENNAI                                 NA
LocationERODE                                   NA
LocationHYDERABAD                               NA
LocationJALANDHAR                               NA
LocationMUMBAI                                  NA
LocationRAIGAD                                  NA

Residual standard error: 0.2141 on 26 degrees of freedom
Multiple R-squared:  0.1196,	Adjusted R-squared:  -0.2867 
F-statistic: 0.2945 on 12 and 26 DF,  p-value: 0.9848
```

The regression model assessing ROA with quadratic Energy Consumption, Carbon Emission, Industry, and Location variables reveals no statistically significant relationships. All predictors, including the

squared term for Energy Consumption, exhibit high p-values, indicating weak individual effects. The model's overall explanatory power is poor, with an adjusted R-squared of -0.2867 and an F-statistic p-value of 0.9848, suggesting that the included variables fail to account for meaningful variation in ROA. Additionally, several location variables are excluded due to multicollinearity, further limiting interpretability. These results imply that neither environmental factors nor categorical industry and location classifications significantly influence asset returns in this dataset.



Despite using quadratic regression with the same ESG predictors, **Return on Assets (ROA)** showed a very poor fit (Adjusted R^2 = –0.296, p = 0.99). This suggests that differences in carbon emissions, energy consumption, industry, and location **do not meaningfully explain profitability relative to total assets** in our sample. ROA is heavily influenced by each firm's asset base—capital-intensive industries naturally report lower ROA regardless of ESG performance—which masks any operational effects. Locations with sparse observations also produced singularities, further weakening the model. While potential improvements (e.g., transforming variables, grouping rare categories, or adding financial controls like leverage) might marginally increase fit, the underlying relationship between ESG metrics and asset efficiency remains weak. **By contrast, Return on Equity (ROE)**—a measure more sensitive to operational profitability—fit strongly (Adjusted R^2 = 0.828, $p < 0.001$). Therefore, **ROE serves as the more reliable profitability indicator for assessing ESG performance growth**, while ROA does not reveal significant ESG-related profitability signals in **THIS** dataset.

4. Conclusion

The regression analyses indicate a clear divergence in how environmental, social, and governance (ESG) factors relate to different profitability measures. Across all specifications, **Return on Assets (ROA)**

consistently exhibits poor explanatory power, with low or negative adjusted R^2 values and non-significant predictors. This suggests that variations in carbon emissions, energy consumption, industry classification, and geographic location do not meaningfully influence firms' asset-based profitability in the sample. The results likely reflect the structural nature of ROA—heavily shaped by asset intensity and capital allocation decisions—which can obscure operational ESG effects.

In contrast, **Return on Equity (ROE)** demonstrates a robust and statistically significant relationship with several ESG variables. Models incorporating quadratic terms and categorical controls achieved a strong fit (Adjusted $R^2 \approx 0.83$, $p < 0.001$), highlighting that **energy consumption has a substantial negative impact on ROE**, while specific industries such as *Textiles and Apparels* and *Paper and Forest Products* show **positive sector-specific advantages**. These findings imply that equity-based profitability is more sensitive to operational efficiency and sectoral ESG practices than asset-based measures.

Overall, **ROE emerges as the more reliable indicator** for evaluating ESG performance growth in this dataset. While modest improvements could be explored—such as variable transformations, merging sparse categories, or adding financial controls like leverage—the evidence suggests that environmental metrics alone are insufficient to explain asset efficiency but are meaningful for equity-based returns. This underscores the importance of selecting appropriate financial performance measures when assessing ESG impacts.